

Tricky stuff

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1 Everything

1.1 Lebesgue-measurable set that is not Borel

Let $\mathcal{C} \subset [0, 1]$ be Cantor set, $c: [0, 1] \rightarrow [0, 1]$ be cantor function defined by

$$\begin{aligned} c &= \frac{1}{2} \text{ on } \left[\frac{1}{3}, \frac{2}{3} \right] \\ c &= \frac{1}{4} \text{ on } \left[\frac{1}{9}, \frac{2}{9} \right], \quad c = \frac{3}{4} \text{ on } \left[\frac{7}{9}, \frac{8}{9} \right] \\ &\vdots \\ c \left(\sum_{n \in \mathbb{N}} \frac{2a_n}{3^n} \right) &= \sum_{n \in \mathbb{N}} \frac{a_n}{2^n} \quad \forall \{a_n\}_{n \in \mathbb{N}}, \quad a_n \in \{0, 1\} \\ c(x) &= \sup_{y \leq x} c(y) \text{ for } x \in [0, 1] \setminus \mathcal{C} \end{aligned}$$

Then c is continuous, surjective and nondecreasing. Therefore

$$\psi(x) := c(x) + x \tag{1}$$

is continuous, increasing and bijective $[0, 1] \rightarrow [0, 2]$. Since

$$\begin{aligned} [0, 1] \setminus \mathcal{C} &= \bigcup_{n \in \mathbb{N}} (a_n, b_n) \text{ where } (a_n, b_n) \text{ are intervals over which } c(x) \text{ is constant} \\ \lambda(\psi(a_n, b_n)) &= \lambda(c(a_n) + a_n, c(b_n) + b_n) = \lambda(c|_{A_n} + a_n, c|_{A_n} + b_n) = \lambda(a_n, b_n) \\ \Rightarrow \lambda(\psi([0, 1] \setminus \mathcal{C})) &= \lambda([0, 1] \setminus \mathcal{C}) = 1 \\ \Rightarrow \lambda(\psi(\mathcal{C})) &= \lambda([0, 2] \setminus \psi([0, 1] \setminus \mathcal{C})) = 1 \end{aligned}$$

By Vitali's theorem, any Lebesgue-measurable set of positive measure has a non-measurable subset. Denote

$$N \subset \psi(\mathcal{C}) \quad (2)$$

such a non-measurable subset. Then $\psi^{-1}(N) \subset \mathcal{C}$ is Lebesgue-measurable since $\lambda(\mathcal{C}) = 0$. But ψ^{-1} is increasing, therefore Borel, so if

$$A := \psi^{-1}(N) \quad (3)$$

was Borel-measurable, also

$$N = (\psi^{-1})^{-1}(A)$$

would be, contradiction. Therefore A is Lebesgue-measurable but not Borel-measurable.

On the other hand, the completion of Borel sigma algebra is easily seen to be Lebesgue sigma algebra:

Denote $\mathcal{B}$ Borel algebra, $\mathcal{L}$ Lebesgue algebra, $\overline{\mathcal{B}}^\lambda$ the completion of $\mathcal{B}$ w.r.t. Lebesgue measure λ , G_δ to be set of countable intersections of open sets ("G_δ" sets). Clearly $\overline{\mathcal{B}}^\lambda \subset \mathcal{L}$. To prove the other inclusion, choose arbitrary $A \in \mathcal{L}$. Then $\exists B \in G_\delta: A \subset B, \lambda(B \setminus A) = 0$, so

$$A = \underbrace{B}_{\in \mathcal{B}} \setminus \underbrace{(B \setminus A)}_{\in \overline{\mathcal{B}}^\lambda, \text{ since } \lambda(B \setminus A) = 0} \in \overline{\mathcal{B}}^\lambda$$

1.2 Composition of Lebesgue measurable functions is not Lebesgue measurable

By definition, $f: \mathbb{R} \rightarrow \mathbb{R}$ is Lebesgue measurable if it is measurable as a map $(\mathbb{R}, \mathcal{L}) \rightarrow (\mathbb{R}, \mathcal{B})$. Let $\psi: [0, 1] \rightarrow [0, 2]$ be as in (1), then since $[0, 1]$ is compact ψ^{-1} is also continuous (therefore Lebesgue measurable). Denote $A \subset [0, 1]$ a non-Borel set from (3), then since $\lambda(A) = 0$, $\mathbb{1}_A: [0, 1] \rightarrow \mathbb{R}$ is Lebesgue measurable. But

$$(\mathbb{1}_A \circ \psi^{-1})^{-1}(\{1\}) = (\psi^{-1})^{-1}(A) = N$$

is the non-measurable set from (2), therefore $\mathbb{1}_A \circ \psi^{-1}$ is not measurable.

Meaning: this is why we define Carathéodory functions.

1.3 Hölder continuous function not absolutely continuous

Let c be the cantor function and c_k be defined by

$$\begin{aligned} c_k \left(\sum_{n=1}^k \frac{2a_n}{3^n} \right) &= \sum_{n=1}^k \frac{a_n}{2^n} \quad \forall \{a_n\}_{n=0}^k, a_n \in \{0, 1\} \\ c_k &\text{ is piecewise linear on } [0, 1] \\ c_k(0) &= 0, c_k(1) = 1 \end{aligned}$$

Then

- $\|c - c_k\|_\infty = 2^{-k}$
- c_k is Lipschitz with constant $\left(\frac{3}{2}\right)^k$ (think about the slope of steepest line)

So for $x, y \in [0, 1]$

$$\begin{aligned} |c(x) - c(y)| &\leq \|c - c_k\|_\infty + |c_k(x) - c_k(y)| + \|c - c_k\|_\infty \\ &\leq 2 \cdot 2^{-k} + \left(\frac{3}{2}\right)^k |x - y| \end{aligned}$$

Choose $k \in \mathbb{N}$ such that $3^{-k-1} \leq |x - y| < 3^{-k}$, then

$$|c(x) - c(y)| \leq 3 \cdot 2^{-k} = 6 \cdot 2^{-k-1} = 6 \cdot 3^{-(k-1) \cdot \log_3 2} < 6 \cdot |x - y|^{\log_3 2}$$

So the cantor function is $\log_3 2$ -Hölder continuous.

Let $\epsilon > 0$. Since c is locally constant on $[0, 1] \setminus \mathcal{C}$, which is a countable union of open intervals with measure one, by continuity of measure we can choose $x_1, \dots, x_n, y_1, \dots, y_n \in [0, 1], x_k < y_k$ (in fact they will be endpoints of form $\frac{l}{3^n}, l, n \in \mathbb{N}$) such that

$$\sum_{k=1}^n |y_k - x_k| > 1 - \epsilon \quad \text{and } c \text{ is constant on intervals } (x_k, y_k)$$

Since c is constant on big set, and $c(0) = 0, c(1) = 1$, it must grow fast on a small set:

$$\begin{aligned} |x_1 - 0| + \sum_{k=1}^{n-1} |x_{k+1} - y_k| + |1 - y_n| &= 1 - \sum_{k=1}^n |y_k - x_k| < \epsilon \\ \text{BUT } c(x_1) - c(0) + \sum_{k=1}^{n-1} (c(x_{k+1}) - c(y_k)) + c(1) - c(y_n) &= 1 - \sum_{k=1}^n (c(y_k) - c(x_k)) = 1 \end{aligned}$$

so c is not $AC[0, 1]$.

A shorter proof might use characterization $AC[0, 1] = W^{1,1}(0, 1)$, since c has classical derivative zero almost everywhere, but certainly

$$c(0) + \int_0^1 c'(x) dx = 0 + \int_0^1 0 dx = 0 \neq c(1)$$

In fact, $c \in BV[0, 1] \setminus AC[0, 1]$ and its Lebesgue-Stieltjes measure is singular nonatomic with respect to the Lebesgue measure on $[0, 1]$.

1.4 Hopf fibration, dirac belt

$$S^1 \rightarrow S^3 \cong B_{1, \mathbb{C}^2}(0) \rightarrow S^2 \cong \mathbb{P}\mathbb{C}^1$$

Hilbert space for two-state system is

$$\mathcal{H} = \mathbb{C}^2 = \mathbb{C} |\uparrow\rangle \oplus \mathbb{C} |\downarrow\rangle$$

When we mod out the norm, we are left with $B_{1, \mathbb{C}^2}(0) \cong S^3$ and when we mod out the phase, we are left with $\frac{S^3}{S^1} \cong S^2$. Quantum evolution happens on S^3 and state space is $S^2 = \mathbb{P}(\mathcal{H})$.

Below we use pauli matrices

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \vec{\sigma} = \begin{pmatrix} \sigma_x \\ \sigma_y \\ \sigma_z \end{pmatrix}$$

The Hopf fibration is the map which sends spinors (elements of S^3) to the expectation values of spin in the x, y, z directions:

$$\begin{aligned} \langle \bullet | \vec{\sigma} | \bullet \rangle : S^3 &\rightarrow S^2 \\ \alpha |\uparrow\rangle + \beta |\downarrow\rangle = |\psi\rangle &\mapsto \langle \psi | \vec{\sigma} | \psi \rangle = \begin{pmatrix} \langle \sigma_x \rangle_\psi \\ \langle \sigma_y \rangle_\psi \\ \langle \sigma_z \rangle_\psi \end{pmatrix} \\ \text{or: } \begin{pmatrix} \alpha \\ \beta \end{pmatrix} &\mapsto \begin{pmatrix} \alpha\bar{\beta} + \beta\bar{\alpha} \\ i(\alpha\bar{\beta} - \beta\bar{\alpha}) \\ |\alpha|^2 - |\beta|^2 \end{pmatrix} \end{aligned}$$

Generally, given a smooth manifold M , a principal $U(1)$ -bundle is a fiber bundle $p: E \rightarrow M$ with a group action $U(1) \times E \rightarrow E$ such that $\forall e \in E, g \in G: e \in p^{-1}(x) \Rightarrow g \cdot e \in p^{-1}(x)$ and the action on each fiber is free and transitive. In our case $M = S^2$.

1.5 Examples of adjunctions

$Unif$ is category of uniform spaces and uniformly cts functions, $CUnif_{T_2}$ is category of complete hausdorff uniform spaces, $\hat{A}$ is completion of A .

$$Hom_{Unif}(A, Y) \cong Hom_{CUnif_{T_2}}(\hat{A}, Y)$$

1.6 Homeomorphism doesn't preserve completeness

Completeness is not a topological invariant. The spaces $(0, 1)$, $\mathbb{R}$ with Euclidean topology are homeomorphic via

$$h: \mathbb{R} \rightarrow (0, 1), \quad h(x) = \frac{\tanh(x) + 1}{2}$$

$\mathbb{R}$ is complete, $(0, 1)$ is not.

1.7 Fréchet space that is not normable

Consider $L_{loc}^1(\mathbb{R}^d)$ given the Fréchet topology generated by seminorms

$$|f|_n = \int_{\|x\| \leq n} |f(y)| dy$$

Suppose there is a norm $\|\bullet\|$ generating the same topology. Then $\exists C > 0, C_n > 0, n \in \mathbb{N}$ such that

$$\|\bullet\| \leq C |\bullet|_N, \tag{4}$$

$$|\bullet|_n \leq C_n \|\bullet\| \tag{5}$$

from this we get $|\bullet|_{N+1} \leq C_{N+1} C |\bullet|_N$, which is clearly wrong for the function $f = \chi_{\{N < \|x\| < N+1\}}$.

To prove (1), for contradiction, assume

$$\forall n, k \in \mathbb{N} \exists \phi_{n,k} \in L_{loc}^1 : \|\phi_{n,k}\| > n |\phi_{n,k}|_k$$

define $\psi_n = \frac{\phi_{n,n}}{n |\phi_{n,n}|_n}$, then

$$\forall l \in \mathbb{N}, n \geq l : |\psi_n|_l \leq |\psi_n|_n = \frac{1}{n} \rightarrow 0 \quad n \rightarrow \infty$$

so $\psi_n \rightarrow 0$ in L_{loc}^1 but $\|\psi_n\| > 1$, contradiction. Generally, for any Fréchet space X where the seminorms are nondecreasing, the inequalities (1) and (2) hold, so if we can find elements $x \in X$ such that $|x|_{N+1}$ can be arbitrarily large and $|x|_N$ arbitrarily small, the space is not normable.

For $C_0^\infty[0, 1]$ where topology is generated by $|f|_n = \sup_{k \leq n} \|f^{(k)}\|_\infty$, the element $f(x) = \frac{1}{n^N} \sin(nx)$ (n large) has small $|\bullet|_{N-1}$ but large $|\bullet|_N$.

1.8 Banach space with no preduals

Suppose $X^* = L^1(0, 1)$. Then by Banach-Alaoglu, $B_X = \{f \in L^1(0, 1) : \|f\| \leq 1\}$ is w^* -compact and convex, so by Krein-Milman $B_X = \overline{\text{aconv}} \text{ ext } B_X$. But there are no extreme points of B_X :

Suppose $f \in B_X$ is extreme. Since $\int_0^x |f(t)|dt$ is continuous in x ,

$$\exists s \in (0, 1) : \int_0^s |f(t)|dt = \frac{\|f\|_1}{2}$$

Now $g_1 := 2f\chi_{(0,s)}$, $g_2 := 2f\chi_{(s,1)}$ satisfy $f = \frac{g_1 + g_2}{2}$, $g_1, g_2 \in B_X$, $g_1, g_2 \neq f$.

1.9 Weak* convergence $\not\Rightarrow$ Weak convergence

Let $X = c_0$, $X^* = l^1$, $X^{**} = l^\infty$. Then the Schauder basis

$$e_n = (0, 0, \dots, 0, \underbrace{1}_{n\text{'th position}}, 0, \dots)$$

satisfies

$$\forall x \in c_0, \langle e_n, x \rangle \rightarrow 0$$

$$\text{For } y = ((-1)^n)_{n=1}^\infty \in l^\infty, \langle y, e_n \rangle = (-1)^n \not\rightarrow 0$$

So $e_n \xrightarrow{w^*} 0$ but $e_n \not\xrightarrow{w} 0$.

1.10 Almost everywhere convergence is not topological

It holds in any topological space (X, τ) and sequence $\{x_n\}_{n \in \mathbb{N}} \subset X$ it holds

$$x_n \xrightarrow{\tau} x \iff \begin{array}{l} \text{For any subsequence } \{x_{n_k}\}_{k \in \mathbb{N}} \text{ of } \{x_n\}, \\ \text{there is a subsequence } \{x_{n_{k_l}}\}_{l \in \mathbb{N}} \text{ such that } x_{n_{k_l}} \xrightarrow{\tau} x. \end{array}$$

Consider the Haar system

$$\begin{aligned} \psi_{n,k} &:= \mathbb{1}_{\left[\frac{k}{2^n}, \frac{k+1}{2^n}\right)}, \quad 0 \leq k \leq 2^n - 1, \quad n \in \mathbb{N}. \\ (\phi_l)_{l \in \mathbb{N}} &:= (\psi_{1,1}, \psi_{2,1}, \psi_{1,2}, \psi_{2,2}, \psi_{3,2}, \psi_{4,2}, \psi_{1,3}, \dots) \end{aligned}$$

Then $\phi \not\xrightarrow{a.e.} 0$, but for any subsequence of $\phi_n = \mathbb{1}_{I_n}$ we can choose a subsequence $\phi_{n_k} = \mathbb{1}_{I_{n_k}}$ such that $\sum_{k \in \mathbb{N}} |I_{n_k}| < \infty$, and therefore from Borel-Cantelli

$$\begin{aligned} \lambda_1(\limsup_{k \rightarrow \infty} I_{n_k}) &= 0 \\ \Rightarrow \lambda_1(\{x \in [0, 1] : x \text{ lies only in finitely many of } I_{n_k}\}) &= 1 \\ \Rightarrow \phi_{n_k} &\xrightarrow{a.e.} 0 \end{aligned}$$

1.11 Symmetric operator with non-real spectrum

Let $A: L^2(0, 1) \rightarrow L^2(0, 1)$ be

$$Au = u'', \quad D(A) = \{u \in W^{2,2}(0, 1): u(0) = u'(0) = u(1) = u'(1) = 0\}.$$

Clearly A is symmetric. It's adjoint is

$$A^*u = u'', \quad D(A^*) = W^{2,2}(0, 1)$$

For any $\lambda \in \mathbb{C}$, equation $A^*u_\lambda = \lambda u_\lambda$ has a two-dimensional space of solutions

$$u_\lambda(x) = C_1 e^{\sqrt{\lambda}x} + C_2 e^{-\sqrt{\lambda}x}$$

And therefore

$$(\text{Im}(\lambda \mathbb{1} - A))^\perp = \text{Ker}(\bar{\lambda} \mathbb{1} - A^*) \neq 0 \Rightarrow \sigma(A) = \mathbb{C}$$

1.12 Fourier series of continuous function does not converge

. Let for $f: [0, 2\pi] \rightarrow \mathbb{R}$

$$S_N f(x) = \sum_{-N}^N a_n e^{-inx} = \frac{1}{2\pi} \int_0^{2\pi} D_N(x-t) f(t) dt,$$

where D_N is the Dirichlet kernel:

$$D_N(x) = \frac{\sin\left(\frac{2N+1}{2}x\right)}{\sin\left(\frac{x}{2}\right)}.$$

It is well-known that

- for $f \in L^p(0, 2\pi)$, $S_N f \rightarrow f$ in $L^p(0, 2\pi)$, $1 < p < \infty$,
- for $f \in BV[0, 1]$, $S_N f \rightarrow f$ pointwise on $[0, 2\pi]$,
- for $f \in AC[0, 1]$, $S_N f \rightrightarrows f$ on $[0, 2\pi]$.

Fix $x \in [0, 2\pi]$, $N \in \mathbb{N}$. Then define a signed Radon measure

$$\Phi_{x,N}: C[0, 2\pi] \rightarrow \mathbb{R}, \quad \Phi_{x,N} f := S_N f(x)$$

it's norm is

$$\|\Phi_{x,N}\|_{(C[0,2\pi])^*} = \frac{1}{2\pi} \int_0^{2\pi} |D_N(s)| dt$$

But

$$\frac{1}{2\pi} \int_0^{2\pi} |D_N(s)| dt \geq \frac{1}{2\pi} \int_0^{2\pi} \frac{|\sin\left(\frac{2N+1}{2}s\right)|}{\frac{s}{2}} ds = \int_0^{(2N+1)\pi} \frac{|\sin x|}{x} dx \xrightarrow{N \rightarrow \infty} \int_0^\infty \frac{|\sin x|}{x} dx = \infty$$

So $\{\Phi_{x,N}\}_{N \in \mathbb{N}}$ is not bounded in $(C[0, 2\pi])^*$. By Uniform Boundedness Principle we get

$$\forall x \in [0, 2\pi] \exists \text{ dense set } \mathcal{A} \subset C[0, 2\pi] \text{ such that } \forall f \in \mathcal{A}: |S_N f(x)| \xrightarrow{N \rightarrow \infty} \infty \quad (6)$$

1.13 Functional not attaining it's norm. Counterexample to projection theorem

Let $X = L^1(0, 1)$, $\phi \in X^*$ defined by $\langle \phi, f \rangle := \int_0^1 xf(x)dx$. We will show

$$\nexists f \in X: \|f\|_X = 1, |\langle \phi, f \rangle| = \|\phi\|_{X^*}.$$

Clearly

$$\|\phi\|_{(L^1(0,1))^*} = \|x\|_{L^\infty(0,1)} = 1.$$

For contradiction, suppose there exist $f \in L^1(0, 1)$ such that $\|f\|_1 = 1$, $|\langle \phi, f \rangle| = 1$. Then f does not change sign on $(0, 1)$ so we can choose f such that $f \geq 0$ on $(0, 1)$ (if not, then $\|f\|_{L^1(0,1)} = \|f\|_{L^1(0,1)} = 1$, $|\langle \phi, |f| \rangle| > |\langle \phi, f \rangle| = 1$, contradicting $\|\phi\|_{(L^1)^*} = 1$). Then

$$\begin{aligned} 1 &= \int_0^{1-\frac{1}{n}} xf(x) dx + \int_{1-\frac{1}{n}}^1 xf(x) dx \leq \\ &\leq \left(1 - \frac{1}{n}\right) \int_0^{1-\frac{1}{n}} f(x) dx + \int_{1-\frac{1}{n}}^1 f(x) dx \\ &= \left(1 - \frac{1}{n}\right) \underbrace{\int_0^1 f(x) dx}_{=1} + \frac{1}{n} \int_{1-\frac{1}{n}}^1 f(x) dx \\ &\Rightarrow \int_{1-\frac{1}{n}}^1 f(x) dx \geq 1 \quad \forall n \in \mathbb{N} \\ &\quad \downarrow n \rightarrow \infty \quad (f \in L^1) \\ &\quad 0 \geq 1. \end{aligned}$$

Intuitively, any f attaining the bound $|\langle \phi, f \rangle| = 1$ would have to have all it's mass at $x = 1$, since if the mass is anywhere else, we can move it closer to $x = 1$ to make $\langle \phi, f \rangle$ bigger. This is impossible on the level of L^1 functions. However, taking $\delta_1 \in (C[0, 1])^*$, $\langle \delta_1, \varphi \rangle = \varphi(1)$, by Hahn-Banach

$$\|\delta_1\|_{(C[0,1])^*} = 1, C[0, 1] \subset L^\infty(0, 1) \Rightarrow \exists T \in (L^\infty(0, 1))^*: T|_{C[0,1]} = \delta_1, \|T\|_{(L^\infty(0,1))^*} = 1.$$

And (equating $\phi \in (L^1)^*$ with $x \in L^\infty$):

$$\langle T, \phi \rangle = \langle T, x \rangle = \langle \delta_1, x \rangle = 1.$$

Therefore ϕ attains it's maximum on the unit ball of dual $(L^\infty)^*$, but not on predual L^1 . The problem is lack of reflexivity. By James' theorem, for any Banach space Y

$$Y \text{ is reflexive} \iff \text{any } \phi \in Y^* \text{ attains it's norm on the unit ball } B_1(0) \subset Y.$$

We say that a subspace $A \subset Y$ of Banach space is *proximal*, if

$$\forall x \in Y \exists! a \in A: \|x - a\| = \inf_{b \in A} \|x - b\| \quad \dots a := P_A(x)$$

by the Projection theorem, every nonempty closed convex subset of a Hilbert space is proximal. We show that in $X = L^1$, $V := \{f \in L^1(0, 1): \langle \phi, f \rangle = 1\}$ is not proximal, even though it is a closed convex subset. We wish to project $0 \in L^1$ to V . Clearly

$$\begin{aligned} \forall \epsilon > 0 \exists f_\epsilon \in L^1: \|f_\epsilon\|_{L^1} \leq 1 + \epsilon, \langle \phi, f_\epsilon \rangle = 1 \quad \& \quad \forall f \in L^1: \langle \phi, f \rangle = 1 \Rightarrow \|f\|_{L^1} \geq 1 \\ &\Rightarrow \inf_{f \in V} \|0 - f\| = \inf_{f \in V} \|f\| = 1 \end{aligned}$$

But we know that this bound cannot be attained, so the projection does not exist:

$$f = P_V(0) \Rightarrow \|f\|_{L^1} = 1 \quad \& \quad f \in V, \quad \text{a contradiction.}$$

1.14 $\int_A \int_B f(x, y) dx dy \neq \int_B \int_A f(x, y) dy dx$

Consider

$$f(x, y) = \frac{x^2 - y^2}{(x^2 + y^2)^2}$$